Reproducing an Analytic Baseline from Bhave & Tonks (2023)

CAMMP | Track A, diffusion-limited corrosion | Mohamede Pankolo | 24 October 2026

What I did

From the Drive folder CAMMP shared, I took the 2023 paper in Ref (Bhave, Zheng, Sridharan & Tonks, "An electrochemical mesoscale tool for modeling the corrosion of structural alloys by molten salt", J. Nucl. Mater. 574 (2023) 154147) and ran the analytic baseline script already provided under workflows/01_analytic_baseline/baseline.py. Before trusting the numbers, I went through the paper itself and checked every constant in the script against the published equations and tables; that part matters more here than just getting the script to run.

The model being reproduced

The paper validates its full 2D/3D phase-field corrosion model (which actually resolves individual grains) against a much simpler closed-form estimate. That estimate is what baseline.py implements:

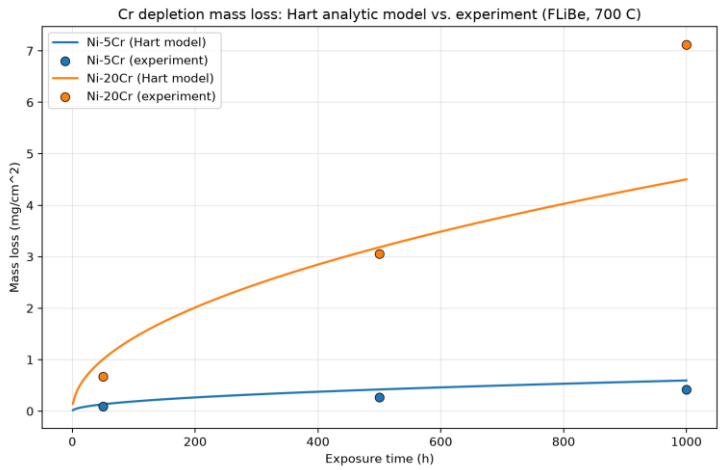
- Eqs. 42-45: Arrhenius diffusivities for Ni and Cr, separately for the grain bulk and the grain boundaries (GB diffusion is always faster).
- Eq. 46 (Hart approximation): combines bulk and GB diffusivity into one effective value, weighted by the fraction of the diffusion path that runs along grain boundaries (dgb / G, with G the average grain size).
- Eq. 50: treats Cr depletion as diffusion into a semi-infinite medium, giving mass loss $\approx 2(c0 - csurf)\cdot\rho\cdot\sqrt{(D\cdot t/\pi)}$.

I checked the constants in the script line by line against the paper: D0 and Q in Eqs. 42-45 match exactly (e.g. Cr grain-boundary diffusion uses D0 = 4.85×10^{10}, Q = 1.962×10^{5} J/mol in both), and csurf = 1.8×10^{-3} and the Table 3 experimental values (density, grain size, mass loss at 50/500/1000 h for Ni-5Cr and Ni-20Cr) are reproduced without any transcription error.

Result

Running baseline.py at T = 973 K (700 °C, the FLiBe exposure temperature used in the paper):

Alloy	t (h)	Exp. (mg/cm²)	Hart (mg/cm²)	Bulk-only	Hart err.
Ni-5Cr	50	0.09	0.13	0.009	+44%
Ni-5Cr	500	0.27	0.42	0.029	+56%
Ni-5Cr	1000	0.42	0.60	0.041	+43%
Ni-20Cr	50	0.67	1.01	0.037	+51%
Ni-20Cr	500	3.06	3.18	0.118	+4%
Ni-20Cr	1000	7.12	4.50	0.166	-37%



Mass loss vs. exposure time: Hart analytic model (lines) against the experimental points from Table 3 (markers).

What the numbers mean

The bulk-only diffusivity is useless on its own: it underestimates mass loss by one to two orders of magnitude (0.166 mg/cm² vs. 7.12 mg/cm² for Ni-20Cr at 1000 h). That confirms the paper's starting premise: Cr depletion in these alloys is dominated by grain-boundary diffusion, not bulk diffusion, which is also why the smaller-grained Ni-20Cr (G = 4.64 μm) corrodes so much faster than Ni-5Cr (G = 16.22 μm) relative to its Cr content.

Folding in the GB contribution through the Hart approximation gets the right order of magnitude and the right qualitative trend across both alloys and all three exposure times. It is not exact, though: for Ni-5Cr it consistently overshoots by 40 to 56%, and for Ni-20Cr it overshoots at 50 h, nearly matches at 500 h, then undershoots by 37% at 1000 h. That inconsistency is expected and is exactly the gap the paper's full model is built to close. Hart's formula assumes grain boundaries run neatly parallel to the diffusion direction and uses one averaged grain size, while the real microstructure has GBs at all angles and a distribution of grain sizes, concentrating depletion unevenly (something the paper shows directly via EBSD-reconstructed 2D simulations in Fig. 5). So this 1D analytic layer is a reasonable sanity check and a cheap way to explore parameter trends, but it is not a substitute for the resolved-grain simulation, which is the point the paper itself makes before introducing the full model.

Where this leaves the rest of the task

The shared workflows folder lays out two further tracks beyond this one:

- A surrogate-model track (Latin-hypercube design + Gaussian-process fit) that mirrors Table 2 of the 2025 coating paper, but it needs real phase-field simulation results to fit against, which means running the MOOSE input files in simulations/. That requires a MOOSE build (via WSL2 on this machine, since MOOSE does not build natively on Windows), which is not set up yet.
- An irradiation-coupled track using Magpie; its own README flags this explicitly as something to check with CAMMP before investing time in, since Bhave & Tonks (2025) say they are already adding radiation effects to the model themselves.

I am holding off on both until I hear back on scope, rather than guessing at several more days of MOOSE setup and simulation work.